Peripheral Interfacing devices

LED

Real world interface: ADC, DAC, SENSORS

Keyboard

LCD



8051 Port Configuration

1) It is suggested to configure the I/O Ports of 8051 before its usage

2) To configure **Port1 as input port**

MOV P1, #0FFH

3) To configure **Port1 as output port**

MOV P1, #00H

4) It need to be done at the beginning of the ASM program



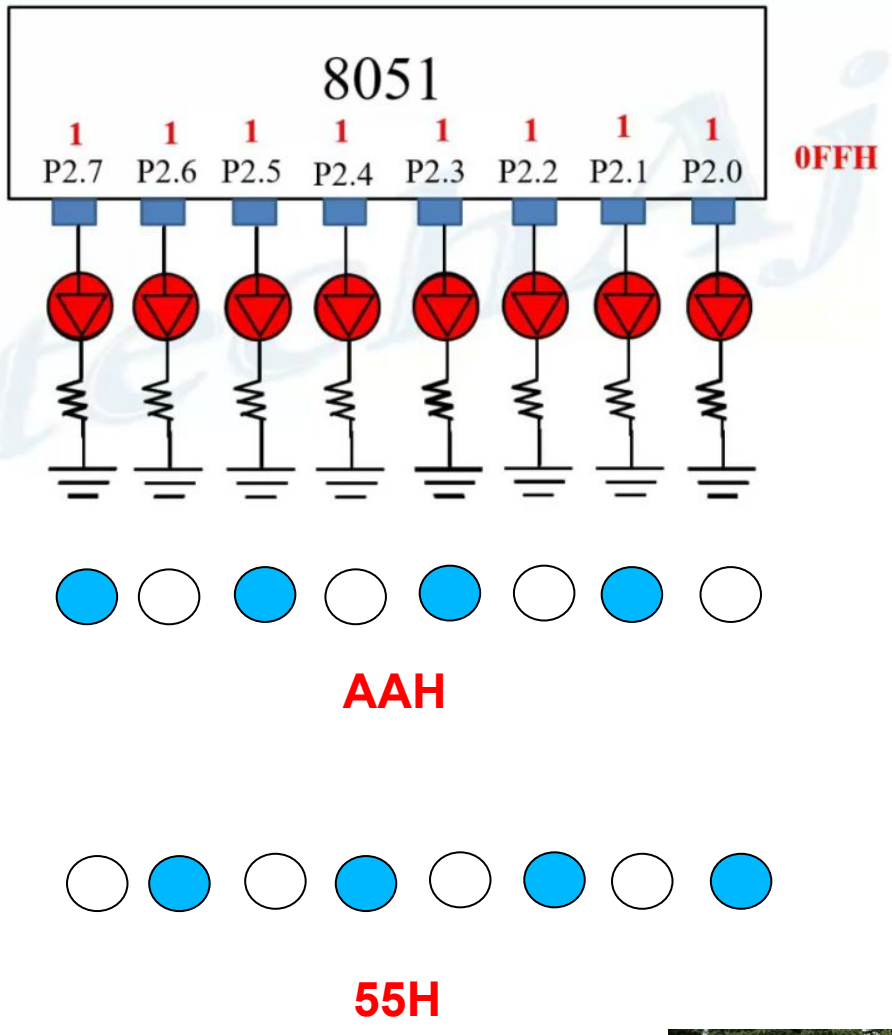
LED interface

Let, alternate LEDS ON/OFF

Send: AAH [10101010]
and 55H [01010101] with some delay

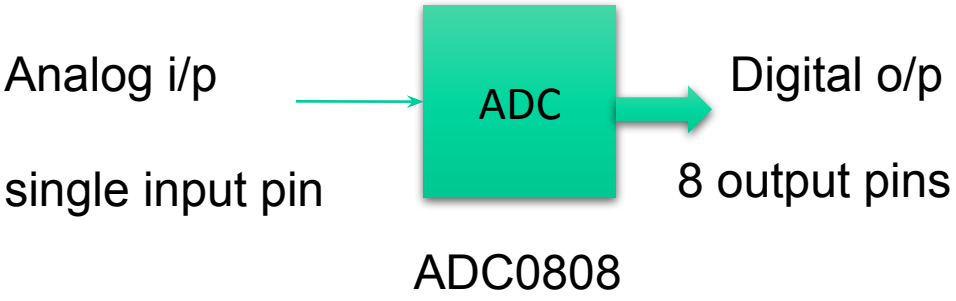
```
        ORG 0000H
L1:     MOV P2, #AAH
        LCALL D1
        MOV P2, #55H
        LCALL D1
        SJMP L1

D1:     DELAY PROGRAM
        WRITE HERE
```





ADC interfacing



8-channel multiplexer. (This ADC can convert 8 different inputs at the same time using the Channel Select pins (CHS0-2)

8-bit ADC (This ADC converts each input into an 8 bit output)

0-5V input rage with 5V power supply.

28-pin DIP (Dual In-Line Package- pins on two side of the IC).

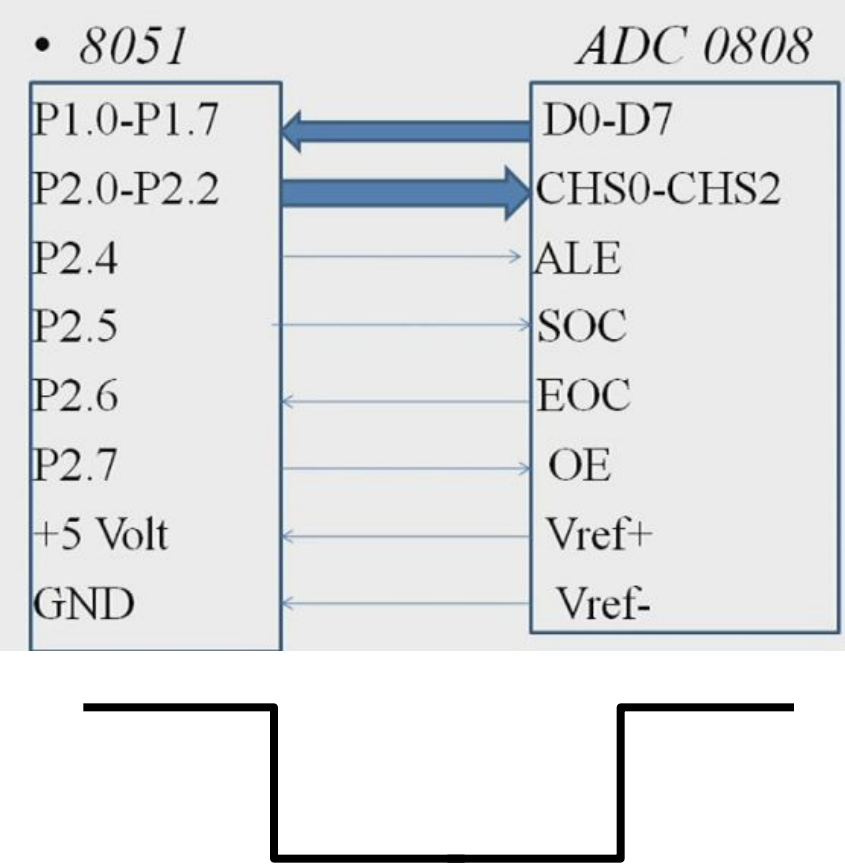
With 5V as Vref(+) and 0V as Vref(-) it provides step size of ~20 mV.(5/256=19.53 mV).



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ADC interface procedure

- **Step 1:** Set the port you connected to the output lines of the ADC as an input port [Port1].
- **Step 2:** Make the Port connected to End of Conversion (EOC) pin high. The reason for doing this is that the ADC sends a high to low signal when the conversion of data is complete
- **Step 3:** ADC channel section will be done by sending 3 bits through P2.0-P2.2
- **Step 4:** Select the data lines according to the input port you want to select. To do this, select the data lines and send a High to Low pulse at the ALE pin to select the address.
- **Step 5:** Now that we have selected the analog input pin, we can tell the ADC to start the conversion by sending a pulse to the (Start of Conversion) SOC pin.
- **Step 6:** Wait for the High to low signal by polling the EOC pin.
- **Step 7:** Wait for the signal to get high again.
- **Step 8:** Extract the converted data by sending a High to low signal to the OE pin (Output Enable).





Interfacing LM35 Temperature Sensor

- Output voltage is linearly proportional to the temperature.
- Sensor gives 10 mV/^{0}C.



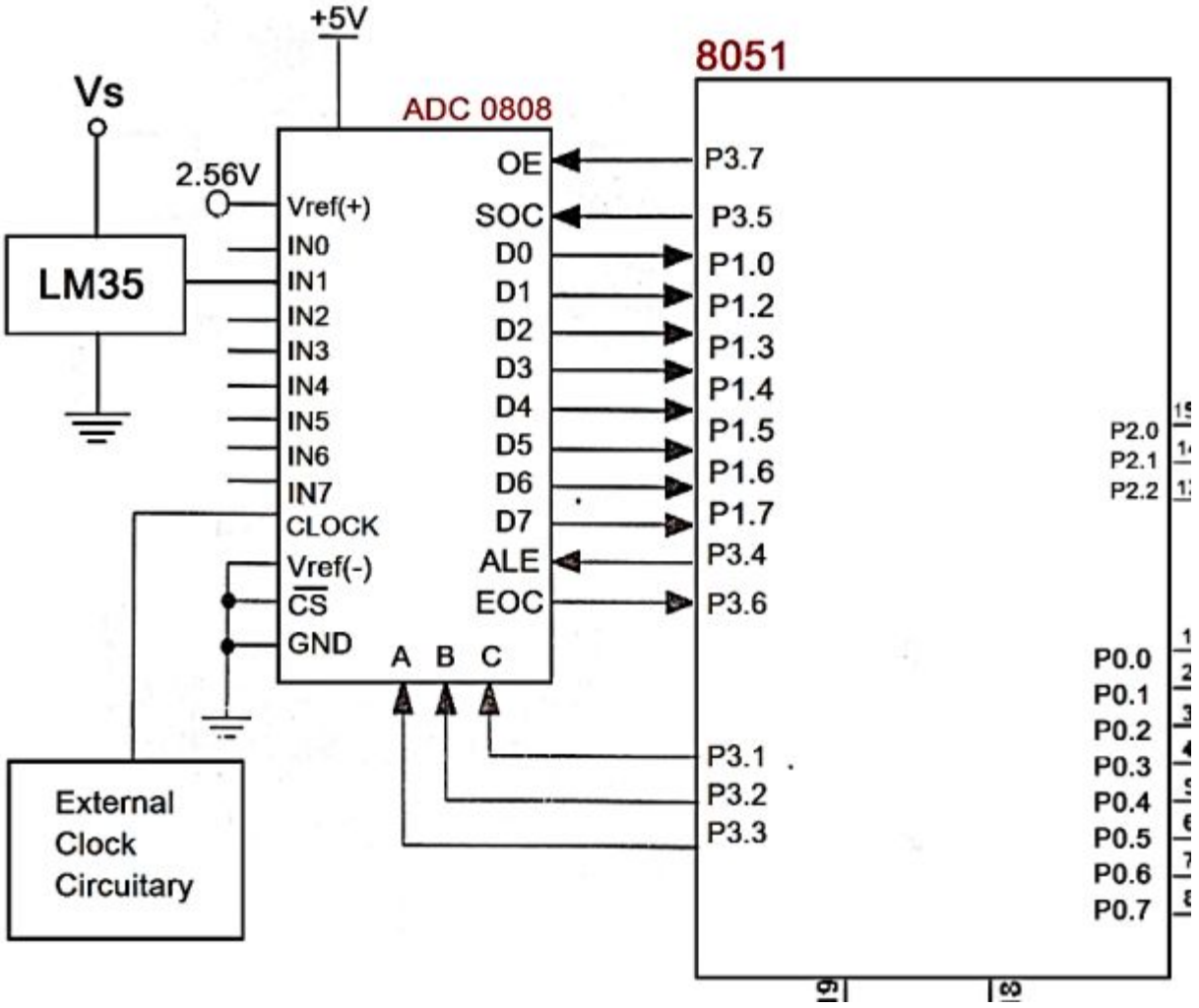
10 mV/^{0}C Step size 10 mV with Vref=2.5V

Temperature Vs. Vout for ADC

Temp (C)	Vin (mV)	Vout (D7-D0)
0	0	0000 0000b (0d)
1	10	0000 0001b (1d)
2	20	0000 0010b (2d)
3	30	0000 0011b (3d)
10	100	0000 1010b (10d)
30	300	0001 1110b (30d)



Interfacing LM35 Temperature Sensor





Interfacing LM35 Temperature Sensor

Component	8051 Connection
LM35 VCC	+5 V
LM35 GND	GND
LM35 VOUT	ADC0808/0809 IN0
ADC D0–D7	Port 1 (P1.0–P1.7)
ADC ALE	P3.3
ADC START	P3.4
ADC EOC	P3.5
ADC OE	P3.6
ADC IN0–IN7 selection	Address = 000
LM35 output	10 mV/°C



Interfacing LM35 Temperature Sensor

- 25°C → LM35 output = **250 mV**
- 30°C → LM35 output = **300 mV**
- 50°C → LM35 output = **500 mV**

Assuming the ADC reference is **5 V** and it is configured for 8-bit conversion:

$$ADC\ Count = \frac{V_{IN}}{5} \times 255$$

Since:

$$V_{LM35} = 10mV/^{\circ}C$$

the temperature can be calculated from the ADC result.



Interfacing LM35 Temperature Sensor

```
ORG 0000H
;-------------------------------------
; LM35 + ADC0808/0809 + 8051
;
; P1      -> ADC D0-D7
; P3.3    -> ALE
; P3.4    -> START
; P3.5    -> EOC
; P3.6    -> OE
; ADC IN0  -> LM35 output
;-------------------------------------
MAIN:
MOV P1, #0FFH      ; Port 1 as input

   CLR P3.3        ; ALE = 0
   CLR P3.4        ; START = 0
   CLR P3.6        ; OE = 0

   ; Select ADC channel IN0
   CLR P3.0
   CLR P3.1
   CLR P3.2
```

```
; Generate ALE pulse
   SETB P3.3
   ACALL DELAY
   CLR P3.3
   ; Generate START pulse
   SETB P3.4
   ACALL DELAY
   CLR P3.4
WAIT_EOC:
   JB P3.5, WAIT_EOC   ; Wait until EOC
becomes LOW
   ; Enable ADC output
   SETB P3.6

   MOV A, P1           ; Read ADC result
   MOV R0, A           ; Store ADC result

   CLR P3.6            ; Disable ADC output
```



Interfacing LM35 Temperature Sensor

```
;-------------------------------
; Temperature calculation
;-------------------------------
; For 5V reference:
   ; Temperature = ADC_value × 500 / 2550
   ;
   ; Approximate calculation:
   ; Temperature = ADC_value × 500 / 255

   MOV A, R0
   MOV B, #500
   MUL AB            ; A:B = ADC × 500

   ; Further division required
   ; depending on ADC reference and
   ; desired accuracy.

   SJMP MAIN
```

```
;-------------------------------------
; Delay
;-------------------------------------
DELAY:
   MOV R6, #10

D1:
   MOV R7, #255

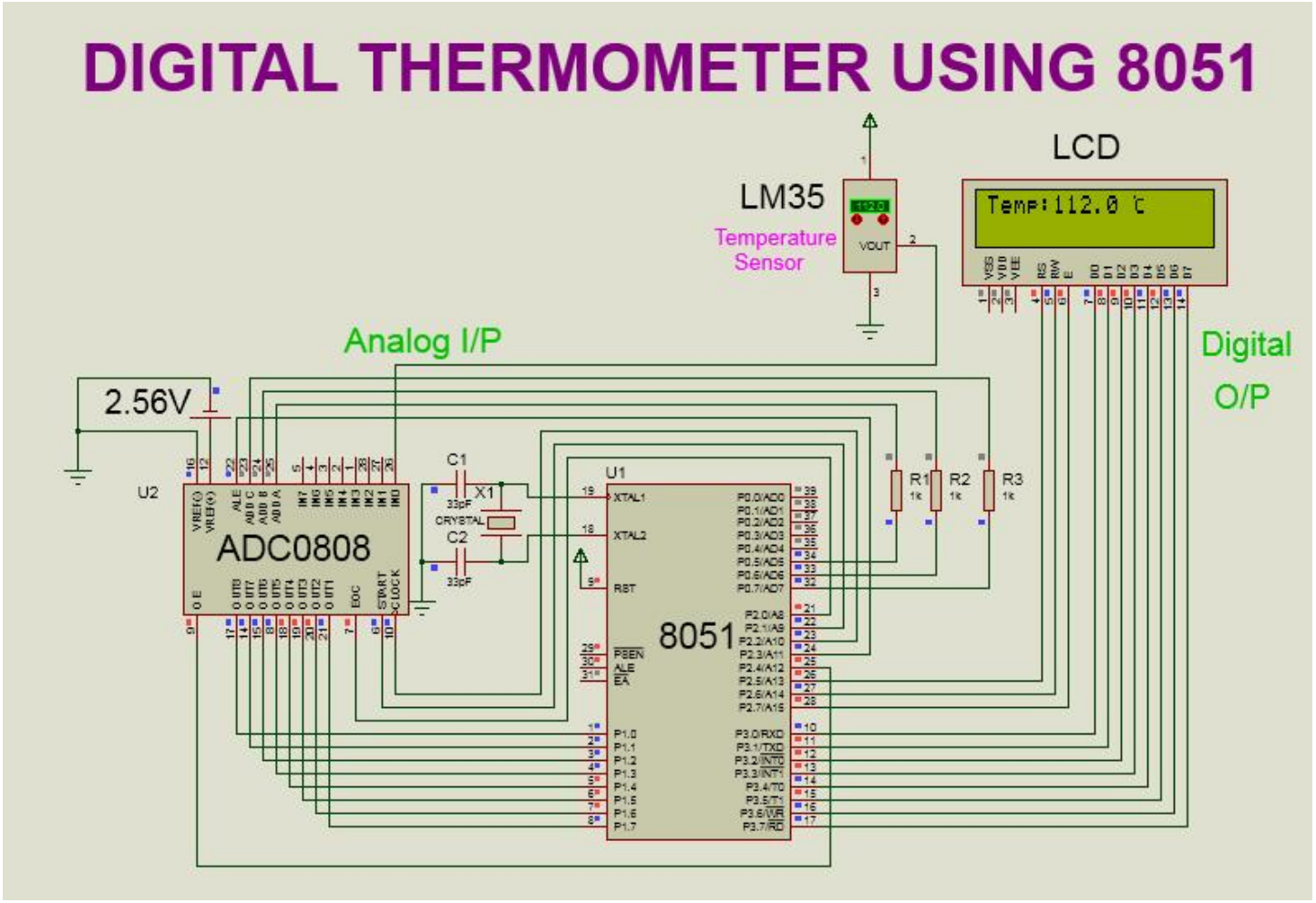
D2:
   DJNZ R7, D2
   DJNZ R6, D1

   RET

END
```



Interfacing LM35 Temperature Sensor





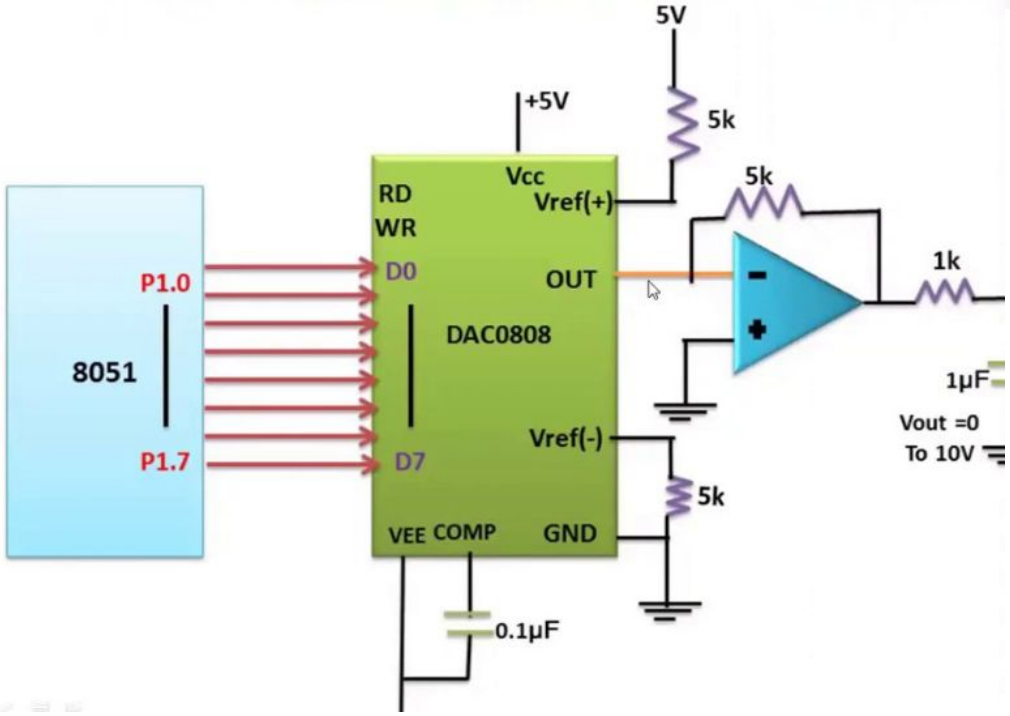
DAC (Digital to Analog Converter) interface

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Procedure for square wave form:

1. Clear P1
2. Call delay
3. Set P1
4. Call delay
5. Repeat from step1



Program for square wave form:

```
        ORG 0000H
REPEAT: MOV P1, #00H
        LCALL D1
        MOV P1, #0FFH
        LCALL D1
        SJMP REPEAT
```

D1: write delay program here as per pulse width

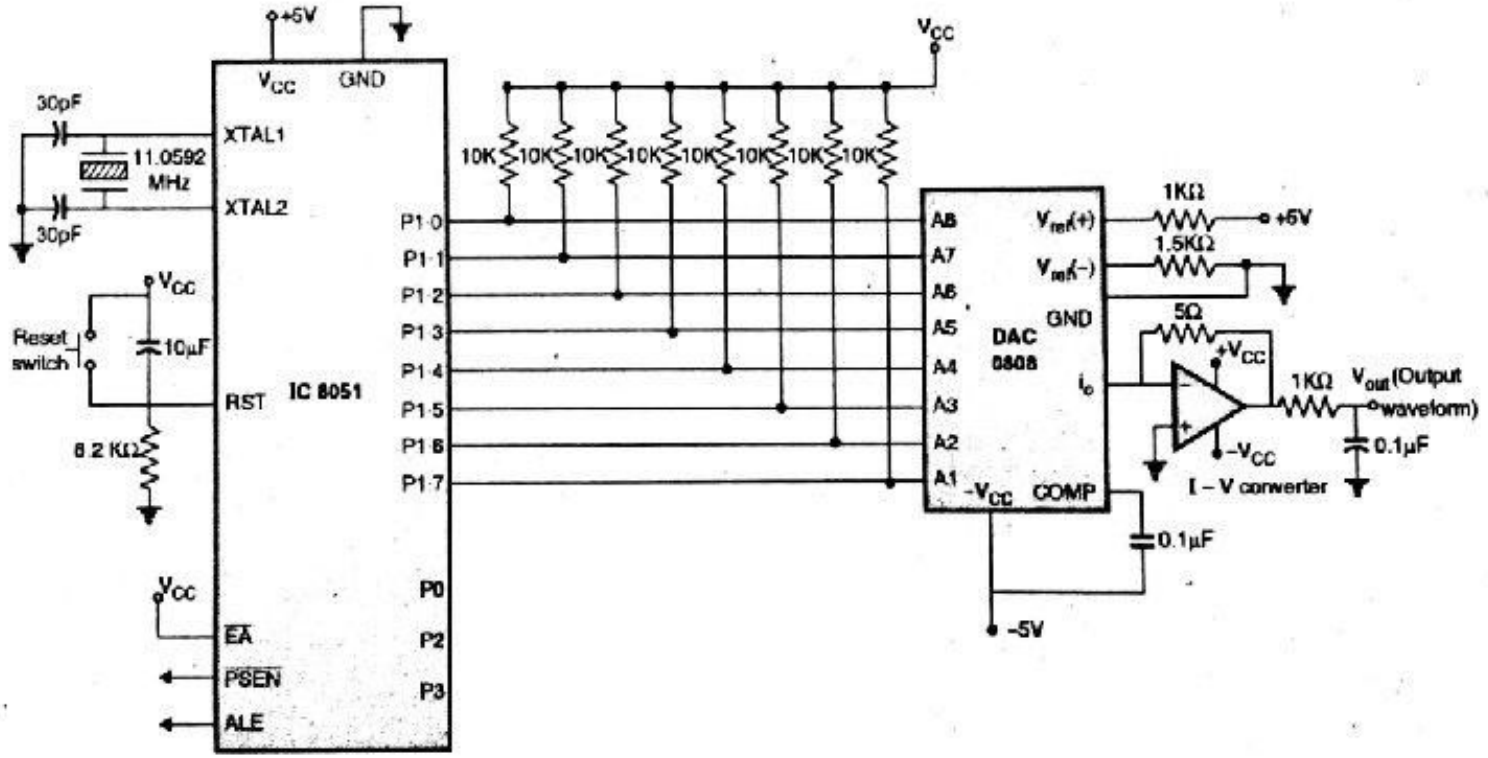
View this for generation of triangular wave form using DAC

https://www.youtube.com/watch?v=xV3lUy0sUHI



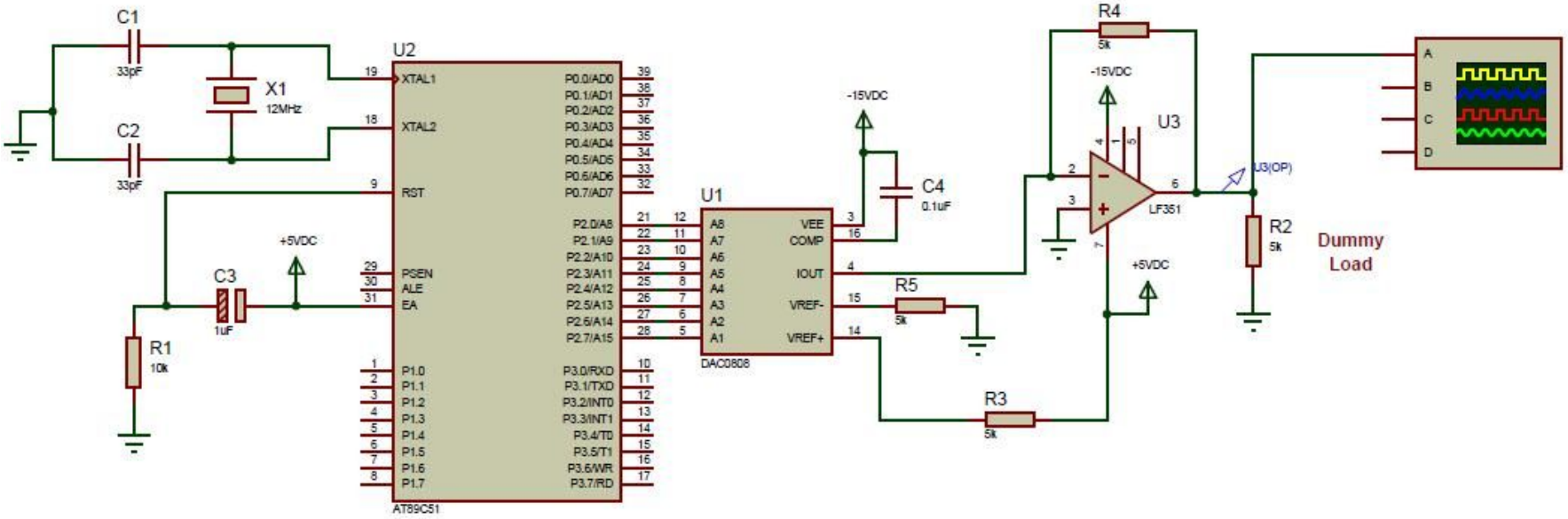
DAC (Digital to Analog Converter) interface

Interfacing

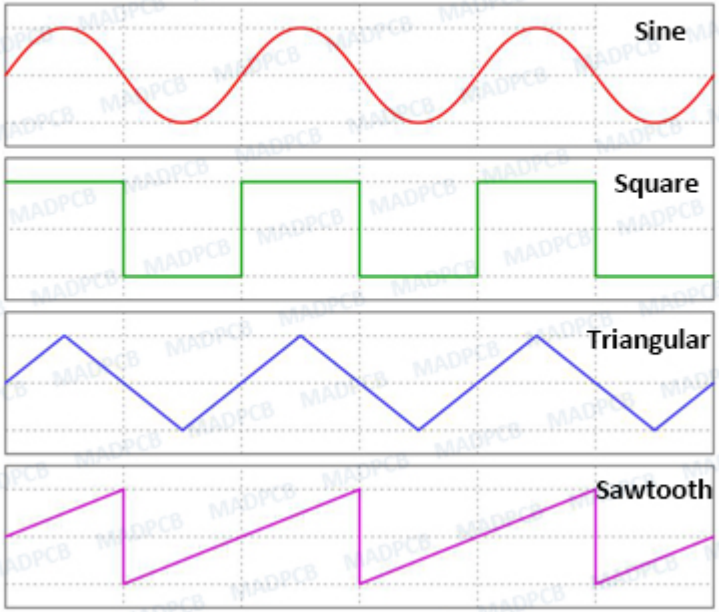




DAC (Digital to Analog Converter) interface



Sine, Square, Triangular, and Sawtooth Waveforms





DAC (Digital to Analog Converter) interface

Sine Wave Generation

```
ORG 0000H
MAIN:
   MOV DPTR,#SINE_TABLE
NEXT:
   CLR A
   MOVC A,@A+DPTR
   MOV P1,A
ACALL DELAY
   INC DPTR
   MOV A,DPL
   CJNE A,#64H,NEXT
   MOV DPTR,#SINE_TABLE
   SJMP NEXT

DELAY:
   MOV R6,#10
D1: MOV R7,#255
D2: DJNZ R7,D2
   DJNZ R6,D1
   RET
END
```

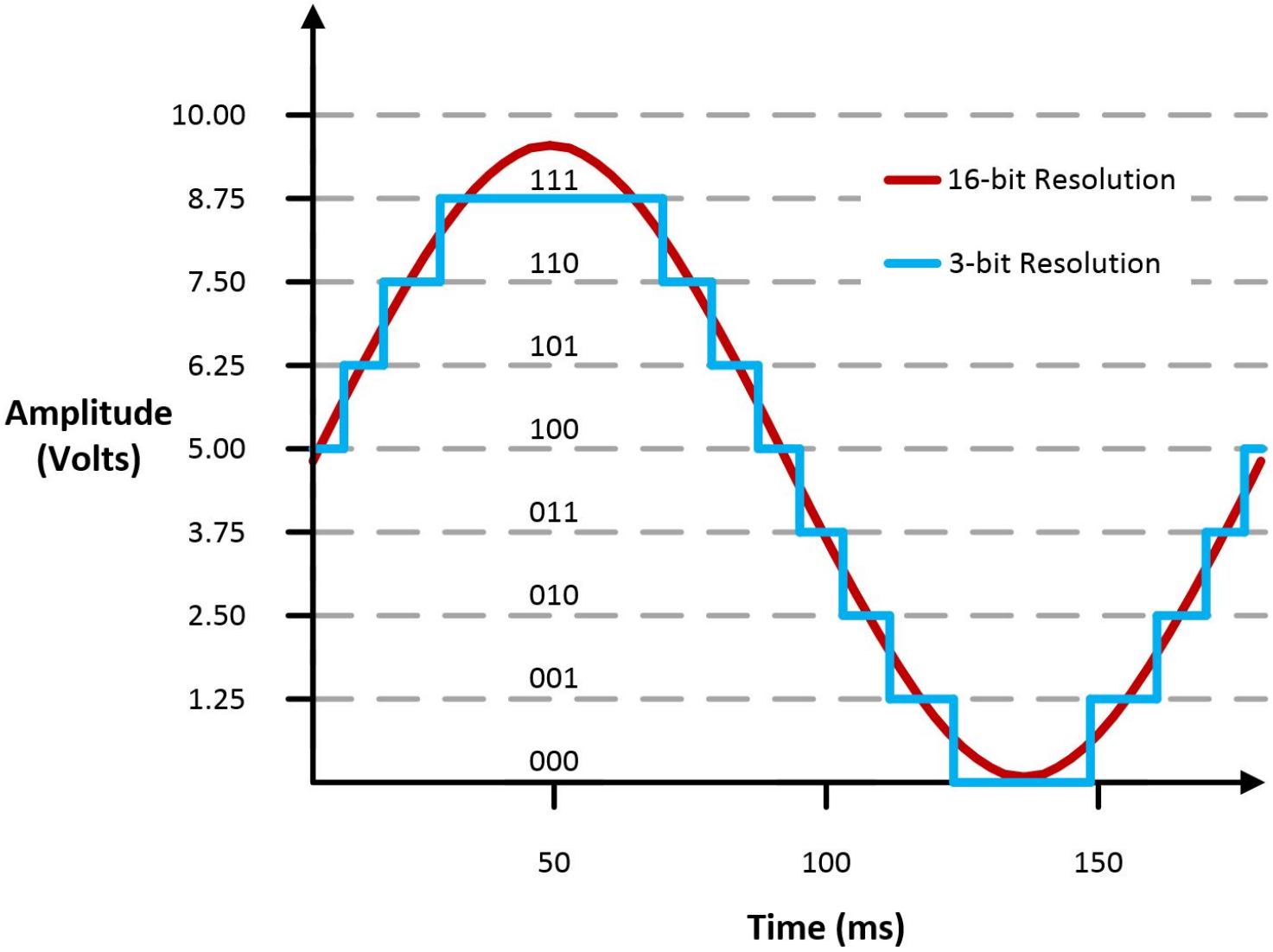
```
;--------------------------------
; Sine wave lookup table
; 100 samples
;--------------------------------
SINE_TABLE:
   DB
128,136,144,152,160,168,176,184,192,199
   DB
206,213,220,226,232,238,244,249,253,255
   DB
255,255,253,249,244,238,232,226,220,213
   DB
206,199,192,184,176,168,160,152,144,136
   DB 128,119,111,103,95,87,79,71,63,56
   DB 49,42,35,29,23,17,11,6,2,0
   DB 0,0,2,6,11,17,23,29,35,42
   DB 49,56,63,71,79,87,95,103,111,119

;--------------------------------
```



DAC (Digital to Analog Converter) interface

Sine Wave Generation





DAC (Digital to Analog Converter) interface

Square Wave Generation

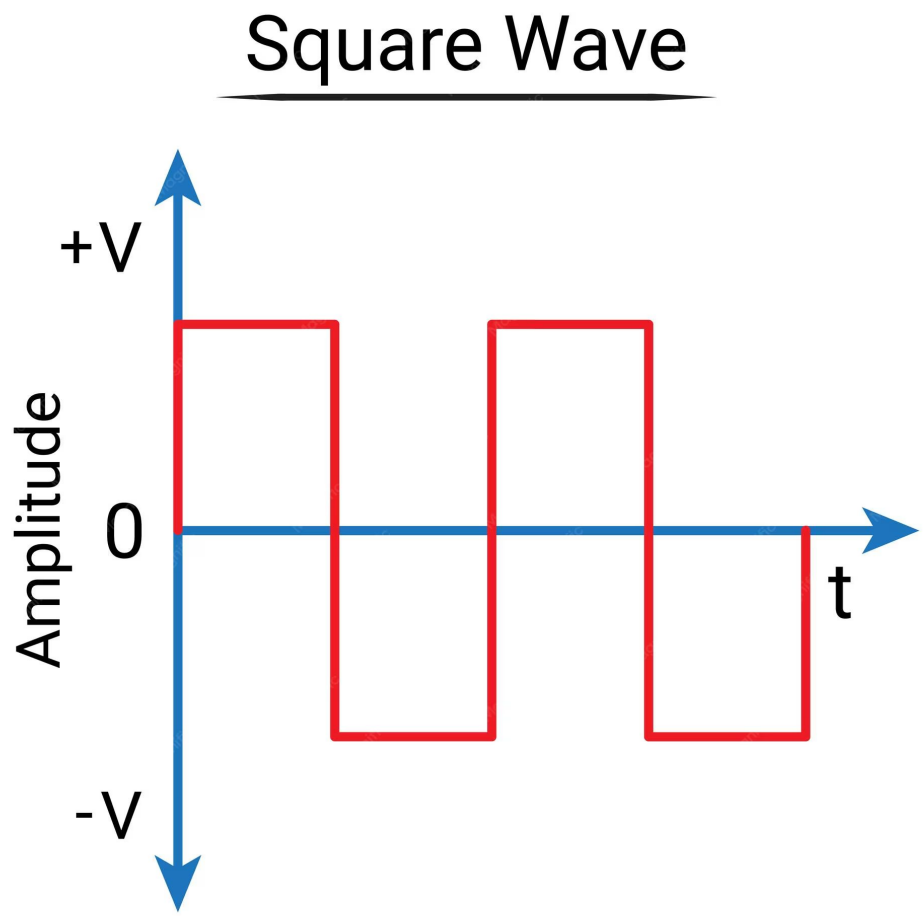
```
ORG 0000H
MAIN:
    MOV P1,#00H
    ACALL DELAY

    MOV P1,#0FFH
    ACALL DELAY

    SJMP MAIN

;-------------------------------
DELAY:
    MOV R6,#20
D1: MOV R7,#255
D2: DJNZ R7,D2
    DJNZ R6,D1
    RET

END
```

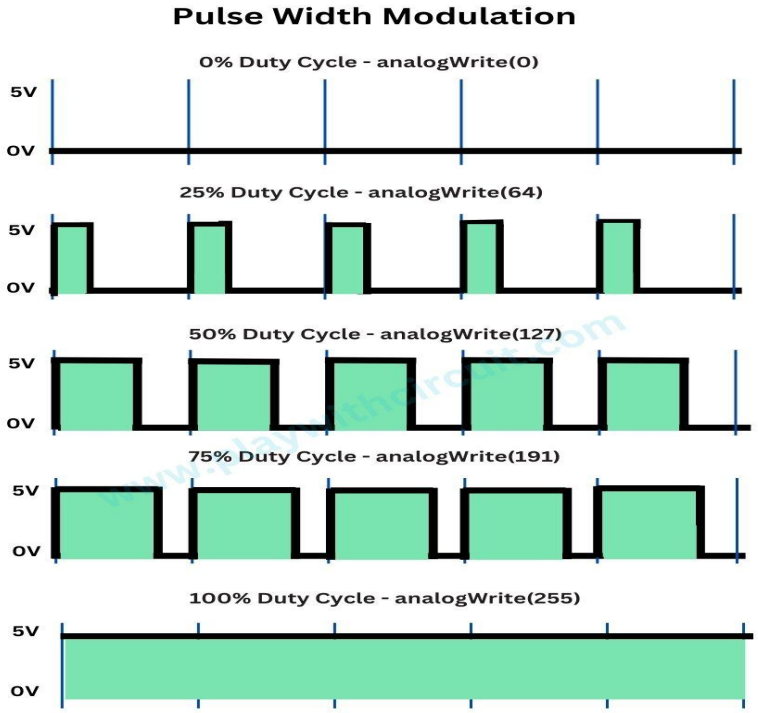




DAC (Digital to Analog Converter) interface

Pulsed Wave Generation

```
ORG 0000H
MAIN:
    ; HIGH pulse
    MOV P1,#0FFH
    ACALL SHORT_DELAY
    ; LOW duration
    MOV P1,#00H
    ACALL LONG_DELAY
    SJMP MAIN
```




```
;--------------------------------
; Short delay
;--------------------------------
SHORT_DELAY:
    MOV R6,#05
S1: MOV R7,#255
S2: DJNZ R7,S2
    DJNZ R6,S1
    RET
;--------------------------------
; Long delay
;--------------------------------
LONG_DELAY:
    MOV R6,#40
L1: MOV R7,#255
L2: DJNZ R7,L2
    DJNZ R6,L1
    RET
END
```



DAC (Digital to Analog Converter) interface

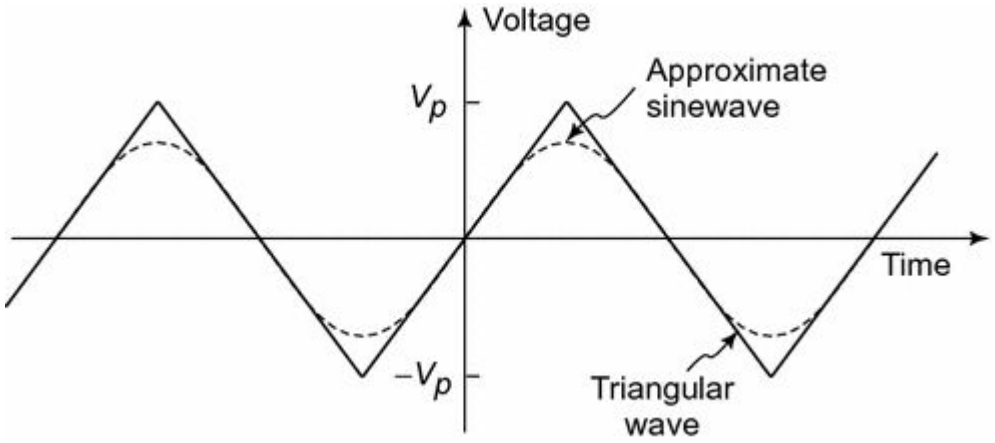
Triangular Wave Generation

```
ORG 0000H
MAIN:
;-------------------------------
; Rising ramp
;-------------------------------
   MOV A,#00H
UP:
   MOV P1,A
   ACALL DELAY

   INC A
   JNZ UP
;-------------------------------
; Falling ramp
;-------------------------------
   MOV A,#0FFH
DOWN:
   MOV P1,A
   ACALL DELAY
   DEC A
   JNZ DOWN
   SJMP MAIN
```

```
;-------------------------------
DELAY:
   MOV R6,#05
D1: MOV R7,#255
D2: DJNZ R7,D2
   DJNZ R6,D1
   RET

END
```





DAC (Digital to Analog Converter) interface

Saw tooth Wave Generation

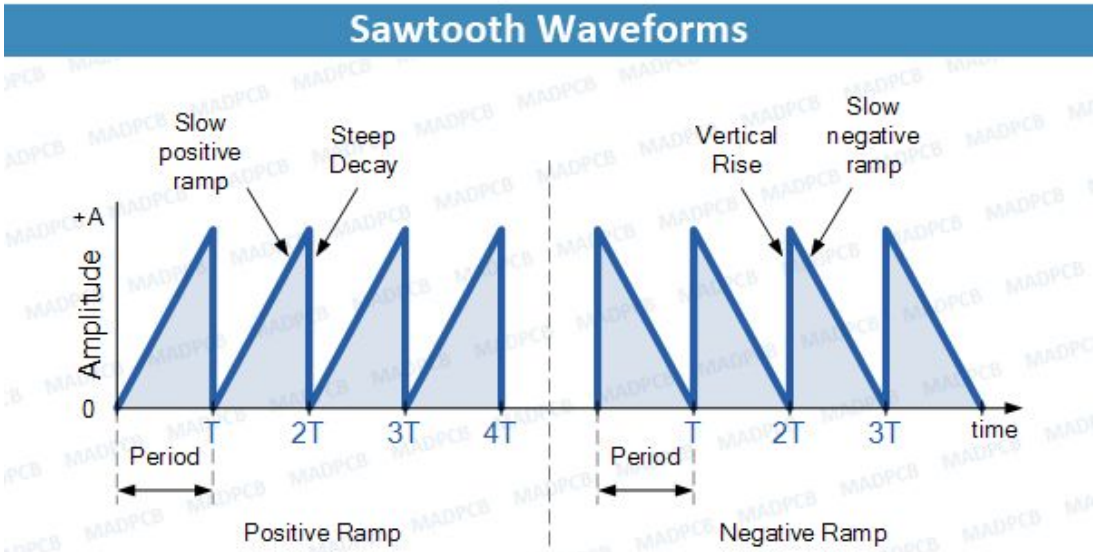
```
ORG 0000H
MAIN:
   MOV A,#00H
UP:
   MOV P1,A          ; Send digital data to DAC
   ACALL DELAY       ; Control frequency

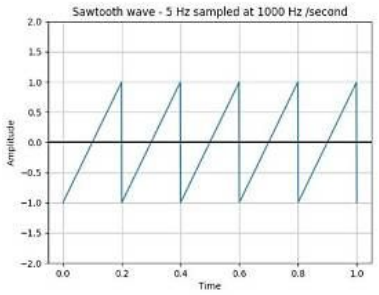
   INC A             ; Increment DAC value
   SJMP UP           ; Continue until A rolls over to
00H
END
```

```
;--------------------------------
; Delay subroutine
;--------------------------------
DELAY:
   MOV R6,#05

D1:
   MOV R7,#255

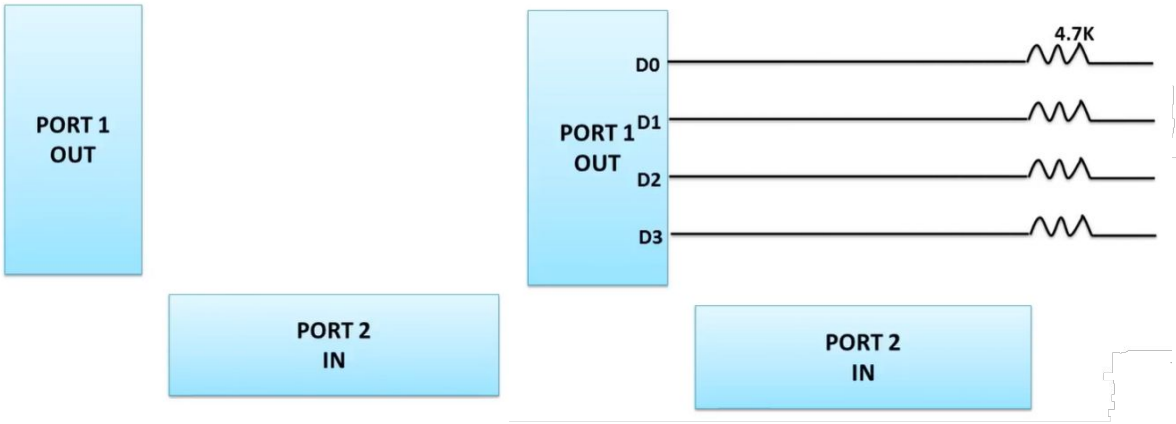
D2:
   DJNZ R7,D2
   DJNZ R6,D1
   RET
```

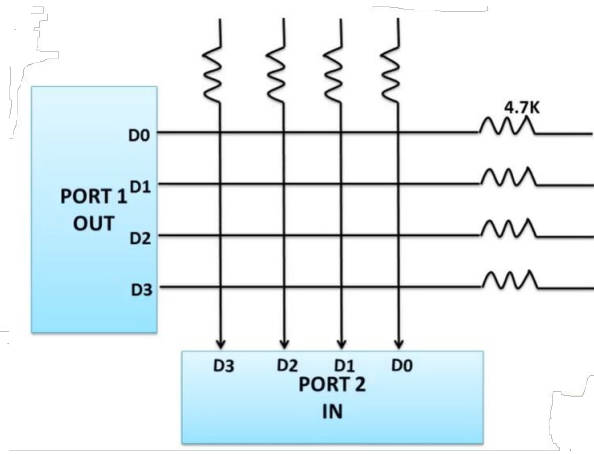


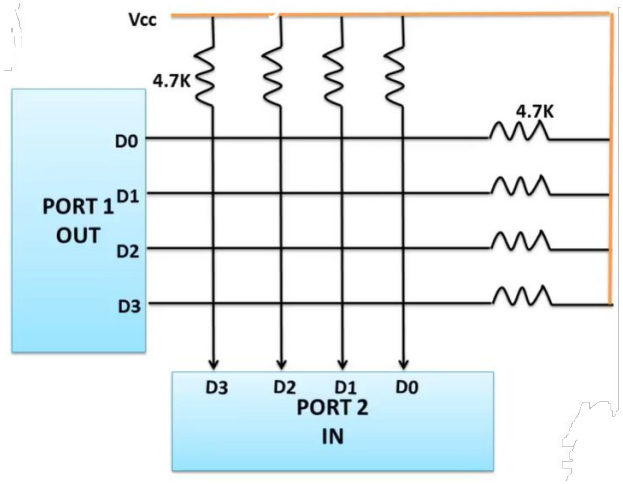




KEYBOARD interface



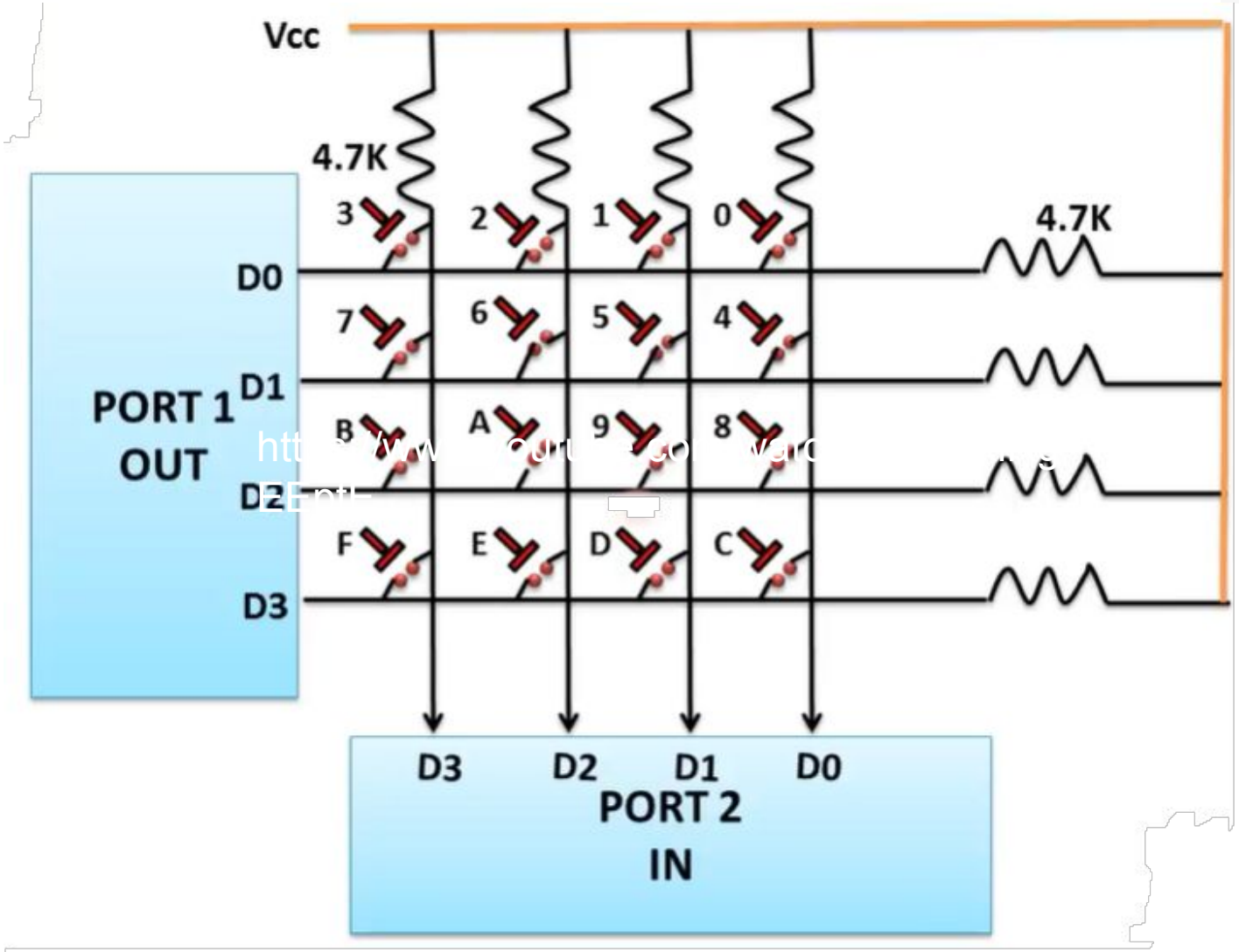




https://www.youtube.com/watch?v=0xHdq0QS1aQ



KEYBOARD interface





KEYBOARD interface

16 keys arranged as a 4X4 matrix

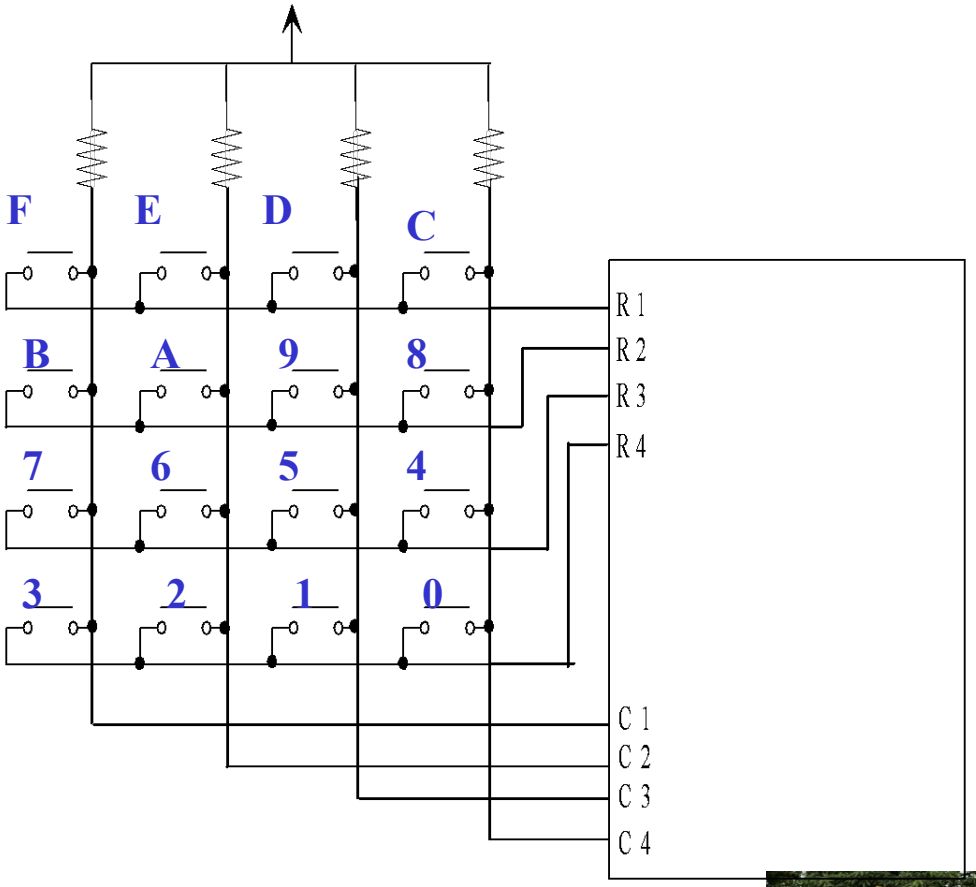
Place a 0 on R0 port

Read C port

If there is a 0 bit
then the button
at the column/row
intersection has
been pressed.

Otherwise, try next row

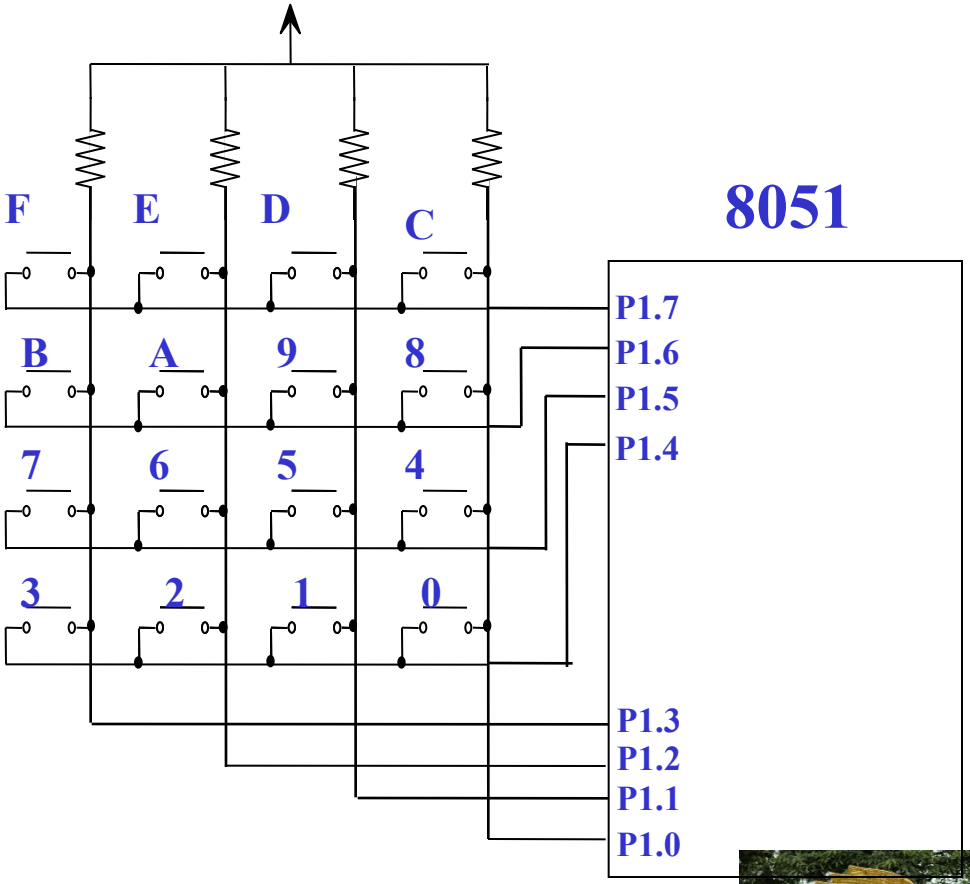
Repeat constantly





KEYBOARD interface

```
scan:   mov P1,#EFH
        jnb P1.0,db_0
scan1: jnb P1.1,db_1
scan2: jnb P1.2,db_2
scan3: jnb P1.3,db_3
scan4: mov P1,#DFH
        jnb P1.0,db_4
……
……
……
```





KEYBOARD interface

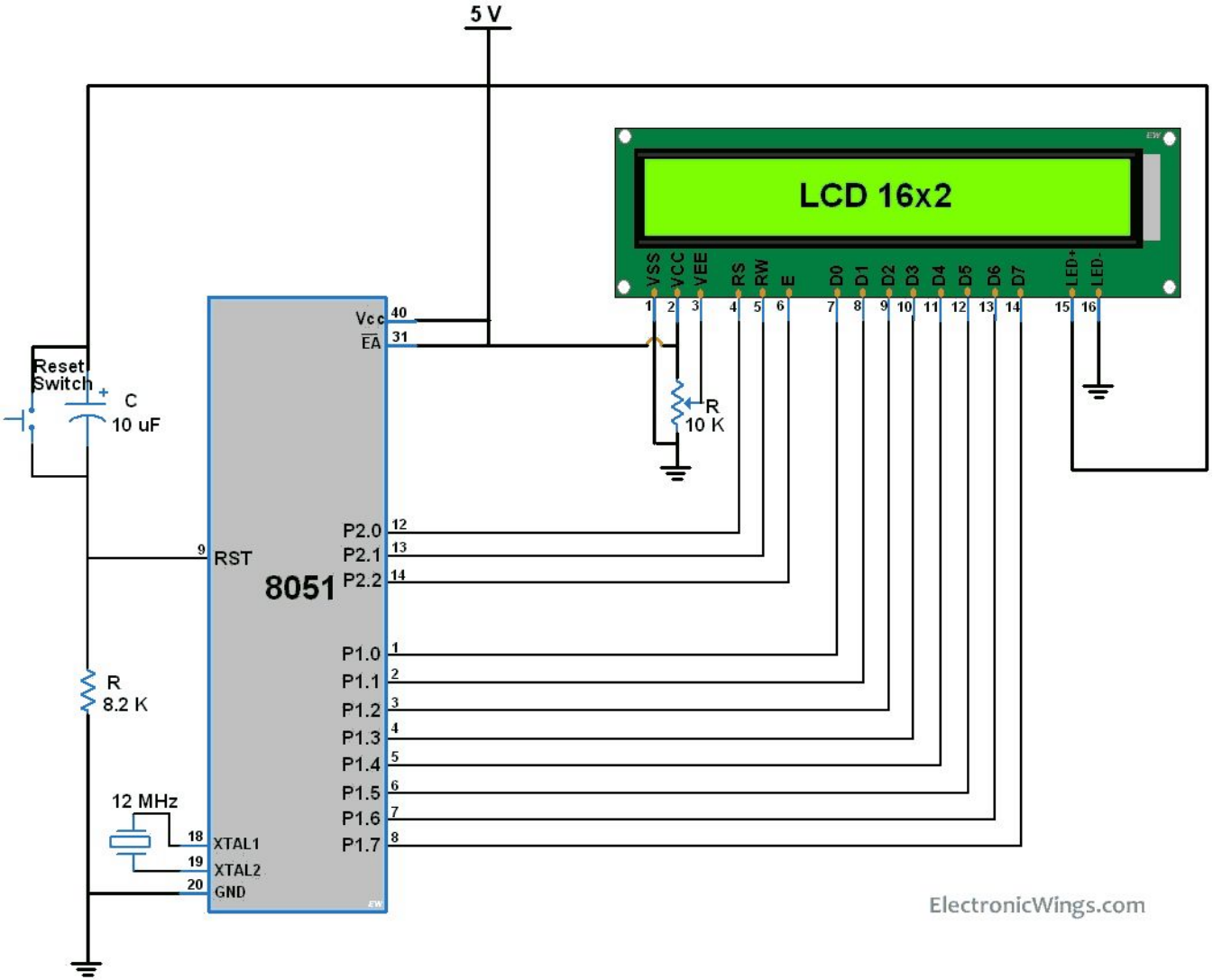
```
db_0:      lcall  wt_10ms
           jb  P1.0, scan1
           movA, #0
           ljmp   get_code
db_1:      lcall  wt_10ms
           jb  P1.1, scan2
           movA, #1
           ljmp   get_code
…..
…
…..
get_code:movDPTR, #key_tab
           movc   A, @A+DPTR
           ljmp   scan
key_tab: db '0123456789ABCDEF'
           END
```



LCD Interfacing

- Liquid Crystal Displays (LCDs)
- cheap and easy way to display text
- Various configurations (1 line by 20 X char up to 8 lines X 80)
- Integrated controller
- The display has two register
  - command register
  - data register
- By RS you can select register
- Data lines (DB7-DB0) used to transfer data and commands

LCD Interfacing



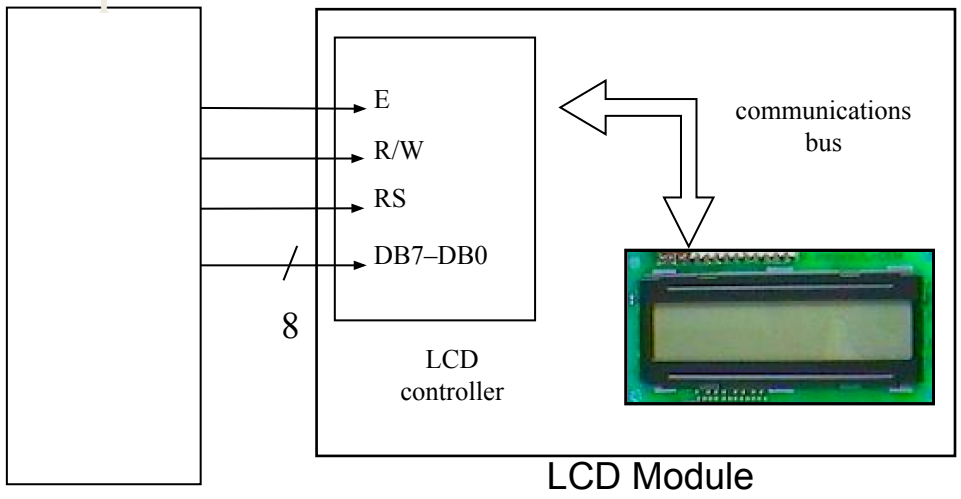
LCD Interfacing

LCD Pin	8051 Connection
D0–D7	P2.0–P2.7
RS	P3.0
RW	P3.1
EN	P3.2
VSS	GND
VDD	+5 V
VEE	Contrast control

LCD Interfacing

Command	Function
38H	8-bit mode, 2-line LCD
0CH	Display ON, cursor OFF
01H	Clear display
06H	Increment cursor
80H	First-line, first position
C0H	Second-line, first position

Alphanumeric LCD Interfacing



- Pinout
  - 8 data pins D7:D0
  - RS: Data or Command Register Select
  - R/W: Read or Write
  - E: Enable (Latch data)

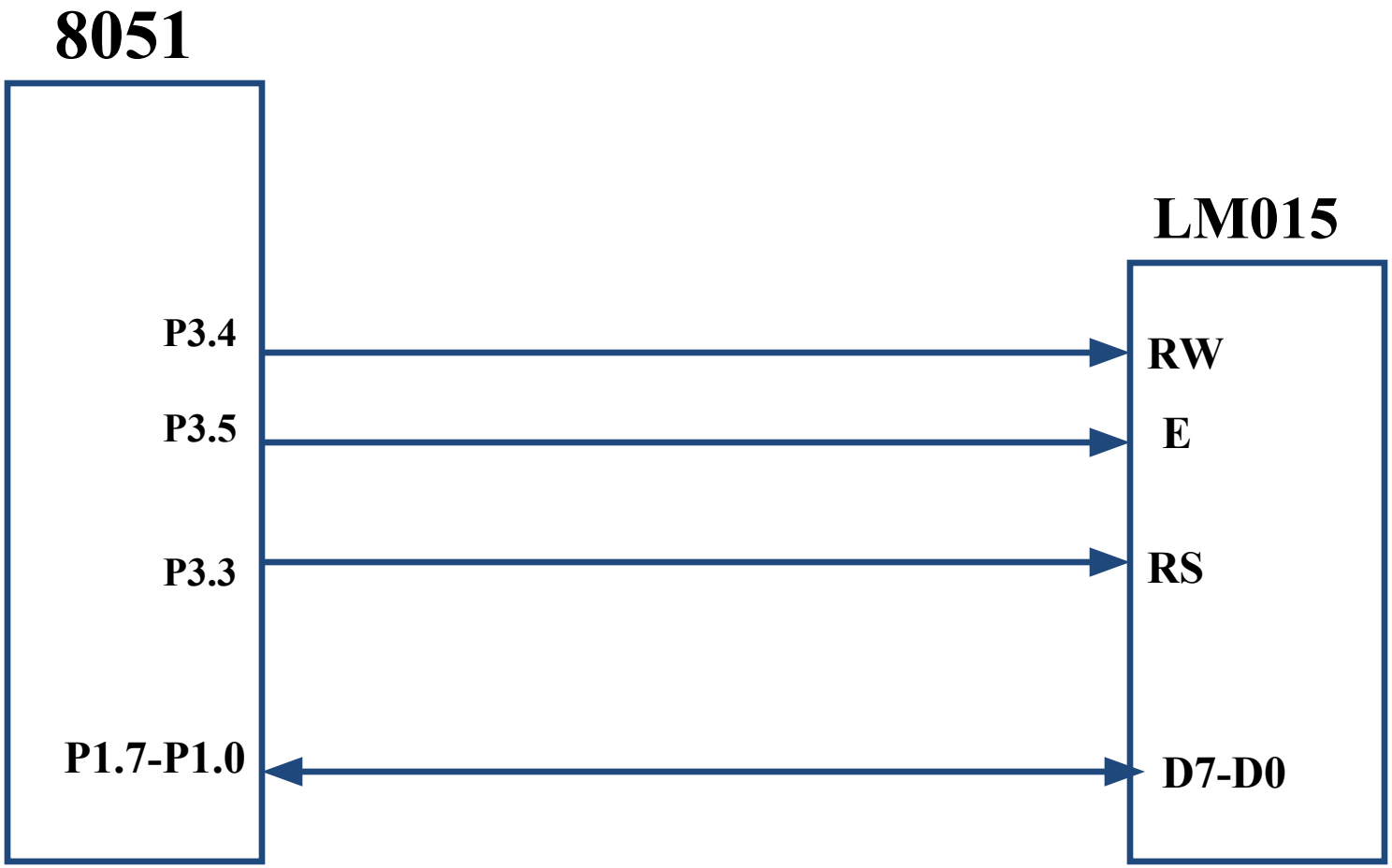
- RS – Register Select
  - RS = 0 → Command Register
  - RS = 1 → Data Register
- R/W = 0 → Write , R/W = 1 → Read
- E – Enable
  - Used to latch the data present on the data pins.

- D0 – D7
  - Bi-directional data/command pins.
  - Alphanumeric characters are sent in ASCII format.

Table 4–10: List of Instructions (Courtesy of Optrex Corporation)

Instruction	Code										Description	Execution Time (max)
	RS	R/W	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0		
Clear Display	0	0	0	0	0	0	0	0	0	1	Clears entire display and sets DD RAM address 0 in address counter.	1.64 ms
Return Home	0	0	0	0	0	0	0	0	1	--	Sets DD RAM address 0 as address counter. Also returns display being shifted to original position. DD RAM contents remain unchanged.	1.64 ms
Entry Mode Set	0	0	0	0	0	0	0	1	I/D	S	Sets cursor move direction and specifies shift of display. These operations are performed during data write and read.	40 µs
Display ON/OFF Control	0	0	0	0	0	0	1	D	C	B	Sets ON/OFF of entire display (D), cursor ON/OFF (C), and blink of cursor position character (b).	40 µs
Cursor or Display Shift	0	0	0	0	0	1	S/C	R/L	--	--	Moves cursor and shifts display without changing DD RAM contents.	40 µs
Function Set	0	0	0	0	1	DL	N	F	--	--	Sets interface data length (DL), number of display lines (L) and character font (F).	40 µs
Set CG RAM Address	0	0	0	1	AGC						Sets CG RAM address. CG RAM data is sent and received after this setting.	40 µs
Set DD RAM Address	0	0	1	ADD							Sets DD RAM address. DD RAM data is sent and received after this setting.	40 µs
Read Busy Flag & Address	0	1	BF	AC							Reads Busy flag (BF) indicating internal operation is being performed and reads address counter contents.	40 µs
Write Data to CG or DD RAM	1	0	Write Data								Writes data into DD RAM or CG RAM.	40 µs
Read Data from CG or DD RAM	1	1	Read Data								Reads data from DD RAM or CG RAM.	40 µs

Interfacing LCD with 8051



Interfacing LCD with 8051

```
ORG 0000H
;-------------------------------------
; LCD Pin Definitions
;-------------------------------------
RS  EQU P3.0
RW  EQU P3.1
EN  EQU P3.2
MAIN:
   ACALL LCD_INIT
   ; Display first line
   MOV A,#80H
   ACALL LCD_CMD
   MOV DPTR,#MSG1
DISPLAY1:
   CLR A
   MOVC A,@A+DPTR
   JZ LINE2
   ACALL LCD_DATA
   INC DPTR
   SJMP DISPLAY1
LINE2:
   ; Move cursor to second line
   MOV A,#0C0H
   ACALL LCD_CMD

   MOV DPTR,#MSG2

DISPLAY2:
   CLR A
   MOVC A,@A+DPTR
   JZ HERE
   ACALL LCD_DATA
   INC DPTR
   SJMP DISPLAY2

HERE:
   SJMP HERE
```

Interfacing LCD with 8051

```
;-------------------------------------
; LCD Initialization
;-------------------------------------
LCD_INIT:
   MOV P2,#00H
   MOV A,#38H      ; 8-bit, 2-line, 5x7 matrix
   ACALL LCD_CMD
   MOV A,#0CH      ; Display ON, cursor OFF
   ACALL LCD_CMD
   MOV A,#01H      ; Clear display
   ACALL LCD_CMD
   MOV A,#06H      ; Increment cursor
   ACALL LCD_CMD
   RET
```

Interfacing LCD with 8051

```
;-------------------------------------
; Send Command to LCD
;-------------------------------------
LCD_CMD:

   MOV P2,A
   CLR RS          ; RS = 0 for command
   CLR RW          ; RW = 0 for write

   SETB EN
   ACALL DELAY
   CLR EN

   ACALL DELAY
   RET
```

```
;-------------------------------------
; Send Data to LCD
;-------------------------------------
LCD_DATA:

   MOV P2,A
   SETB RS         ; RS = 1 for data
   CLR RW          ; RW = 0 for write

   SETB EN
   ACALL DELAY
   CLR EN

   ACALL DELAY
   RET
```

Interfacing LCD with 8051

```
;-------------------------------------
; Delay
;-------------------------------------
DELAY:

   MOV R6,#20

D1:
   MOV R7,#255

D2:
   DJNZ R7,D2
   DJNZ R6,D1

   RET
```

```
;-------------------------------------
; Messages
;-------------------------------------
MSG1:
   DB 'VIT BHOPAL',00H

MSG2:
   DB 'Dream to Win',00H

END
```

Interfacing LCD with 8051

